SHARP COEFFICIENTWISE POSITIVITY FOR A MATRIX TURÁN DETERMINANT OF ${}_0F_1$

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Abstract. We prove a sharp coefficientwise positivity theorem for a matrix Turán determinant of the reciprocal-gamma normalization $Z(a,\lambda) = {}_0F_1(;a;\lambda)/\Gamma(a) = \sum_{k\geq 0} \lambda^k/(k!\,\Gamma(a+k))$, $a > 0$. From parameter and Euler derivatives of Z, we form a one-parameter family of symmetric 2×2 matrices $\mathcal{T}_\kappa(a, \lambda)$ and prove that every positive-degree Maclaurin coefficient of $\Delta^{(\kappa)} = \det \mathcal{T}_\kappa$ is strictly positive for all $a > 0$ if and only if $\kappa \geq 1$. Thus $\kappa = 1$ is the sharp uniform deformation threshold, and $\Delta^{(1)}(a, \cdot)$ is strictly absolutely monotone. At this endpoint, coefficientwise determinant positivity persists although the first nonconstant coefficient matrix may be indefinite: a mixed-determinant polarization confines its possible negative contribution to the degree-two coefficient, whose positivity follows from an explicit decomposition. Via the identity $I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a, \lambda)$, an exact diagonal congruence transfers the endpoint theorem to the modified Bessel function I_ν, yielding, for every $\nu > -1$, a positive-definite Bessel–Schur matrix coupling order, argument, and mixed order–argument derivatives of $\log I_\nu$. Both the uniform deformation threshold and, for each fixed ν, the least argument-independent correction are sharp.

1. Introduction

Turán inequalities for modified Bessel functions are closely connected with order monotonicity, logarithmic derivatives, and total positivity; see [1, 2, 4, 21]. A complementary literature proves scalar log-concavity and coefficientwise Turán inequalities for reciprocal-gamma and hypergeometric series [9, 10, 11, 13]. Here we study the normalized series

$$Z(a, \lambda) = \sum_{k\geq 0} \frac{\lambda^k}{k!\Gamma(a + k)}$$

through a coupled 2×2 matrix involving parameter and Euler derivatives.

A single theorem underlies the paper: the sharp coefficientwise classification of Section 2. The Bessel inequality is its corollary, while the optimality analysis determines the exact κ-threshold and the least z-independent correction on the positive-series domain $a > 0$ (equivalently $\nu > -1$)—the range in which every coefficient of Z is positive—and the negative-order analysis exhibits an explicit regime where coefficientwise positivity of Δ fails. The associated Bessel–Schur matrix couples three second-order quantities—the concavity of $\log I_\nu$ in the order, the curvature in the argument, and the mixed order–argument sensitivity—and Δ is the corresponding differential Schur complement. By a *matrix Turán determinant* we mean this confluent Turánian assembled from derivatives, rather than a determinant of parameter-shifted evaluations (compared in Section 8).

The proof proceeds in several stages. We begin in Section 3 with an asymmetric Chu–Vandermonde convolution, from which we derive exact formulas for the coefficient matrices of $\mathcal{T}_\kappa$. These formulas expose the structure of the sharp endpoint $\kappa = 1$: the associated finite convolution law singles out $m = 1$ as the potentially exceptional positive degree.

In Section 4, explicit ℓ^2 Gram representations prove that every coefficient matrix of degree $m \geq 2$ is positive definite, whereas M_1 can be indefinite for small $a > 0$. We then pass in

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Section 5 from positivity of the individual coefficient matrices to positivity of the determinant coefficients by polarizing the determinant. The potentially indefinite matrix M_1 can contribute negatively only through its self-pairing, which occurs solely in the coefficient Δ_2; an independent explicit decomposition proves that this exceptional coefficient is nevertheless strictly positive. This completes the endpoint theorem, and the full sharp classification follows by comparing $\mathcal{T}_\kappa$ with $\mathcal{T}_1$.

Finally, in Sections 6 and 7, an explicit diagonal congruence transports the endpoint theorem to a positive-definite Bessel–Schur matrix and a mixed order–argument inequality for every $\nu > -1$. The same reduction determines the exact uniform deformation threshold and, for each fixed order, the least argument-independent correction. We also examine the boundary of the positive-series domain: although the leading small-argument term remains positive away from the poles, the continued determinant already loses coefficientwise positivity on $-3 < \nu < -2$.

2. Main results

Throughout, $a > 0$, $\lambda \geq 0$, and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. We use

$$Z(a,\lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!\Gamma(a+k)} = \frac{{}_0F_1(;a;\lambda)}{\Gamma(a)}. \tag{2.1}$$

The series in (2.1) and all fixed-order a- and Euler derivatives used below converge locally uniformly on $\{\Re a > 0\} \times \mathbb{C}$: the polygamma factors arising from $\partial_a^j \left[1/\Gamma(a+k)\right]$ grow at most polynomial–logarithmically in k, so on compact sets each differentiated series is dominated by a constant multiple of $e^{|\lambda|}$ and termwise differentiation is valid. Put $\Theta_\lambda = \lambda\partial_\lambda$ and $g = \psi_1(a)$, where $\psi = \psi^{(0)}$ and $\psi_1 = \psi^{(1)}$ are the digamma and trigamma functions. In the Bessel variable we use the distinct notation $\Theta_z = z\partial_z$. Subscripts denote application of the indicated operator, so $Z_a = \partial_a Z$, $Z_{\Theta_\lambda} = \Theta_\lambda Z$, and $Z_{a\Theta_\lambda} = \partial_a \Theta_\lambda Z$. For $\kappa \in \mathbb{R}$, define

$$A = Z_a^2 - ZZ_{aa}, \tag{2.2}$$

$$B = Z^2 + ZZ_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda}, \tag{2.3}$$

$$C_\kappa = Z^2 + g\big(\kappa Z Z_{\Theta_\lambda} - ZZ_{\Theta_\lambda\Theta_\lambda} + Z_{\Theta_\lambda}^2\big), \tag{2.4}$$

$$\mathcal{T}_\kappa = \begin{pmatrix} A & \sqrt{g}\,B \\ \sqrt{g}\,B & C_\kappa \end{pmatrix}, \qquad \Delta^{(\kappa)} = \det \mathcal{T}_\kappa = AC_\kappa - gB^2. \tag{2.5}$$

At $\kappa = 1$ write $C = C_1$, $\mathcal{T} = \mathcal{T}_1$, and $\Delta = \Delta^{(1)}$.

Remark 2.1 (Canonical origin of the determinant)**.** Writing $L = \log Z$, direct differentiation gives

$$\frac{A}{Z^2} = -L_{aa}, \qquad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \qquad \frac{C_\kappa}{gZ^2} = \frac{1}{g} + \kappa L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}.$$

Thus the entries are logarithmic curvatures of Z in the parameter and Euler directions, with κ deforming the argument-direction term. At the endpoint $\kappa = 1$, the normalization is chosen so that the identity (6.1) and a diagonal congruence carry $\mathcal{T}$ to the order–argument Bessel–Schur matrix of Theorem 2.4; see Section 6.

A function on $(0,\infty)$ is *strictly absolutely monotone* if it and all of its derivatives are strictly positive there. Here *uniform* means that a single threshold in κ works for every $a > 0$; it does not assert a positive lower bound uniform in a.

Theorem 2.2 (Sharp coefficientwise classification)**.** *Uniform coefficientwise positivity holds exactly for $\kappa \geq 1$: for every $a > 0$ and $\kappa \geq 1$,*

$$\Delta^{(\kappa)}(a,\lambda) = \sum_{n\geq 1} \Delta_n^{(\kappa)}(a)\lambda^n, \qquad \Delta_n^{(\kappa)}(a) > 0, \tag{2.6}$$

whereas for $\kappa < 1$ the linear coefficient is negative for all sufficiently small $a > 0$. Consequently, $\Delta(a,\cdot)$ is strictly absolutely monotone on $(0,\infty)$ and $\mathcal{T}(a,\lambda) \succ 0$ for $a, \lambda > 0$.

The series in (2.6) starts at degree one because

$$A(a,0)=\frac{g}{\Gamma(a)^2},\qquad B(a,0)=C_\kappa(a,0)=\frac{1}{\Gamma(a)^2},$$

so $\mathcal{T}_\kappa(a,0)=\Gamma(a)^{-2}\left(\begin{smallmatrix} g & \sqrt{g}\\ \sqrt{g} & 1\end{smallmatrix}\right)$ has rank one and $\Delta^{(\kappa)}(a,0)=0$.

Remark 2.3 (Rank-one boundary and the sharp threshold)**.** The rank-one constant term is shared by the entire family $\mathcal{T}_\kappa$, so the sign of the first nonzero determinant coefficient governs the onset of coefficientwise positivity. The linear coefficient of $\Delta^{(\kappa)}$ is negative for all sufficiently small $a>0$ whenever $\kappa<1$, whereas Theorem 2.2 proves positivity of every positive-degree coefficient for $\kappa\ge 1$. Thus $\kappa=1$ is the exact uniform deformation threshold.

For $\nu>-1$ and $z>0$, the identity (6.1) and the positive coefficients of $Z(a,\lambda)$ show that $I_\nu(z)>0$. Thus the following real logarithmic derivatives are well defined:

$$G_\nu=-\partial_\nu^2\log I_\nu, \tag{2.7}$$

$$P_\nu=\partial_\nu(\Theta_z\log I_\nu), \tag{2.8}$$

$$H_\nu=2(\Theta_z\log I_\nu-\nu)-\Theta_z^2\log I_\nu, \tag{2.9}$$

and

$$D_\nu=G_\nu\left(H_\nu+\frac{4}{\psi_1(\nu+1)}\right)-(1+P_\nu)^2. \tag{2.10}$$

The shifts in $1+P_\nu$ and $4/\psi_1(\nu+1)$ are exactly those produced by the diagonal congruence (6.5). Here $P_\nu=1+2L_{a\Theta_\lambda}$, so $1+P_\nu=2(1+L_{a\Theta_\lambda})$; as $z\downarrow 0$ one has $L_{a\Theta_\lambda}\to 0$, hence $P_\nu\to 1$ and $1+P_\nu\to 2$. For $\kappa\in\mathbb{R}$, define the deformation

$$H_\nu^{(\kappa)}=2\kappa(\Theta_z\log I_\nu-\nu)-\Theta_z^2\log I_\nu, \tag{2.11}$$

$$D_\nu^{(\kappa)}=G_\nu\left(H_\nu^{(\kappa)}+\frac{4}{\psi_1(\nu+1)}\right)-(1+P_\nu)^2. \tag{2.12}$$

At $\kappa=1$ these reduce to H_ν and D_ν. We call this the *fixed-correction deformation* because the z-independent correction $4/\psi_1(\nu+1)$ is held fixed while the argument-curvature term is deformed.

Theorem 2.4 (Mixed Bessel–Schur inequality)**.** *For $\nu>-1$ and $z>0$, $D_\nu(z)>0$, equivalently*

$$G_\nu(z)\left(H_\nu(z)+\frac{4}{\psi_1(\nu+1)}\right)>(1+P_\nu(z))^2. \tag{2.13}$$

Corollary 2.5 (Positive Bessel–Schur matrix)**.** *For $\nu>-1$ and $z>0$,*

$$\begin{pmatrix} G_\nu(z) & 1+P_\nu(z)\\ 1+P_\nu(z) & H_\nu(z)+4/\psi_1(\nu+1)\end{pmatrix}\succ 0.$$

As $z\downarrow 0$, the matrix-valued function admits a continuous extension whose value is the rank-one matrix $\left(\begin{smallmatrix} g & 2\\ 2 & 4/g\end{smallmatrix}\right)$, where $g=\psi_1(\nu+1)$.

Proposition 2.6 (Sharpness of the Bessel correction and deformation)**.** *For each fixed $\nu>-1$, $4/\psi_1(\nu+1)$ is the least real constant R such that*

$$G_\nu(z)\big(H_\nu(z)+R\big)>(1+P_\nu(z))^2\qquad \text{for every } z>0.$$

Moreover, for the fixed-correction deformation $D_\nu^{(\kappa)}$ defined in (2.12)*,*

$$D_\nu^{(\kappa)}(z)>0\qquad \text{for every } \nu>-1 \text{ and } z>0$$

holds if and only if $\kappa\ge 1$.

Thus the correction is sharp separately for each fixed ν, whereas the deformation threshold is sharp uniformly over the full domain $\{(\nu,z):\nu>-1,\ z>0\}$.

3. Reciprocal-gamma convolution and coefficient formulas

The coefficient calculation rests on a two-parameter Chu–Vandermonde identity.

Lemma 3.1 (Asymmetric reciprocal-gamma convolution)**.** *Let $m \in \mathbb{N}_0$ and let $\alpha, \beta > 0$. Then*

$$\sum_{i=0}^{m} \frac{1}{i!(m-i)!\Gamma(\alpha+i)\Gamma(\beta+m-i)} = \frac{(\alpha+\beta+m-1)_m}{m!\Gamma(\alpha+m)\Gamma(\beta+m)}. \tag{3.1}$$

Both sides extend meromorphically in (α, β).

Proof. Multiply the left-hand side by $m!\Gamma(\alpha+m)\Gamma(\beta+m)$. Using

$$\frac{\Gamma(\alpha+m)}{\Gamma(\alpha+i)} = (\alpha+i)_{m-i} = \frac{(\alpha)_m}{(\alpha)_i}$$

and

$$\frac{\Gamma(\beta+m)}{\Gamma(\beta+m-i)} = (\beta+m-i)_i = (-1)^i(1-\beta-m)_i,$$

the resulting sum is

$$(\alpha)_m {}_2F_1\left(\begin{matrix}-m, 1-\beta-m \\ \alpha\end{matrix}; 1\right).$$

Chu–Vandermonde gives $(\alpha+\beta+m-1)_m$. ∎

Set

$$S_m = S_m(a) = \frac{(2a+m-1)_m}{m!\Gamma(a+m)^2} > 0.$$

Taking $\alpha = \beta = a$ in Lemma 3.1 gives

$$[\lambda^m]Z(a,\lambda)^2 = S_m.$$

Theorem 3.2 (Exact coefficient matrices)**.** *There are expansions*

$$A = \sum_{m\geq 0} S_m \alpha_m \lambda^m, \qquad B = \sum_{m\geq 0} S_m \beta_m \lambda^m, \qquad C_\kappa = \sum_{m\geq 0} S_m \gamma_m^{(\kappa)} \lambda^m, \tag{3.2}$$

where

$$\alpha_m = \psi_1(a+m), \tag{3.3}$$

$$\beta_0 = 1, \qquad \beta_m = \frac{2a+m-2}{2(a+m-1)} \quad (m \geq 1), \tag{3.4}$$

and

$$\gamma_m^{(\kappa)} = 1 + g c_m^{(\kappa)}, \tag{3.5}$$

with

$$c_0^{(\kappa)} = 0, \qquad c_1^{(\kappa)} = \frac{\kappa-1}{2}, \tag{3.6}$$

and, for $m \geq 2$,

$$c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{m(2a+m-2)}{2(2a+2m-3)} = \frac{m\left[(\kappa-1)(2a+2m-3)+m-1\right]}{2(2a+2m-3)}. \tag{3.7}$$

At the sharp endpoint $\kappa = 1$,

$$c_0 = c_1 = 0, \qquad c_m = \frac{m(m-1)}{2(2a+2m-3)} \quad (m \geq 2).$$

Consequently,

$$\mathcal{T}(a,\lambda) = \sum_{m=0}^{\infty} S_m M_m \lambda^m, \qquad M_m = \begin{pmatrix} \alpha_m & \sqrt{g}\,\beta_m \\ \sqrt{g}\,\beta_m & 1+gc_m \end{pmatrix}. \tag{3.8}$$

The theorem does not assert $M_m \succeq 0$ for every m: M_1 can be indefinite for small $a > 0$ (see Section 4). For general κ, write $M_m^{(\kappa)}$ for the coefficient matrix of $\mathcal{T}_\kappa$, obtained by replacing $1 + gc_m$ with $1 + gc_m^{(\kappa)}$.

Proof. For the first coefficient, define

$$F_m(\delta) = \sum_{i=0}^{m} \frac{1}{i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)}.$$

By Lemma 3.1,

$$F_m(\delta) = \frac{(2a+m-1)_m}{m!\Gamma(a+m+\delta)\Gamma(a+m-\delta)}. \tag{3.9}$$

On the one hand,

$$\partial_\delta^2 \big[Z(a+\delta,\lambda)Z(a-\delta,\lambda)\big]\big|_{\delta=0} = 2ZZ_{aa} - 2Z_a^2 = -2A. \tag{3.10}$$

On the other hand, $F_m'(0) = 0$ and the logarithmic second derivative of (3.9) is $-2\psi_1(a+m)$ at $\delta = 0$, so

$$F_m''(0) = -2S_m\psi_1(a+m). \tag{3.11}$$

Comparing the coefficient of λ^m in (3.10) with (3.11) gives $[\lambda^m]A = S_m\psi_1(a+m)$, proving (3.3).

For $m = 0$, one has $[\lambda^0]B = Z(a,0)^2 = S_0$, so $\beta_0 = 1$. Assume henceforth that $m \ge 1$. Let I be the index of the first factor in $Z(a+\delta,\lambda)Z(a-\delta,\lambda) = \sum_m F_m(\delta)\lambda^m$, so that $[\lambda^m]$ weights i by

$$w_i(\delta) = \big[i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)\big]^{-1}.$$

For real δ with $|\delta| < a$, write $\mathbb{E}_\delta$ for expectation under the probability weights $w_i(\delta)/F_m(\delta)$, and put $\mathbb{E} = \mathbb{E}_0$, $\mathsf{D} = m - 2I$, and $J = m - I$. Applying Lemma 3.1 to $\sum_i i\, w_i(\delta)$, with $(\alpha,\beta) = (a+1+\delta, a-\delta)$ and $m-1$ in place of m, gives

$$\sum_{i=0}^{m} i\, w_i(\delta) = \frac{(2a+m-1)_{m-1}}{(m-1)!\,\Gamma(a+m+\delta)\Gamma(a+m-1-\delta)}.$$

Dividing by (3.9) yields

$$\mathbb{E}_\delta I = \frac{m(a+m-1-\delta)}{2(a+m-1)},$$

whence

$$\mathbb{E}_\delta \mathsf{D} = \frac{m\delta}{a+m-1}. \tag{3.12}$$

The score $\partial_\delta \log w_i(\delta)\big|_{\delta=0}$ is

$$X = \psi(a+m-I) - \psi(a+I),$$

so differentiating (3.12) at $\delta = 0$ gives $\mathbb{E}(\mathsf{D}X) = \partial_\delta \mathbb{E}_\delta \mathsf{D}|_{\delta=0}$, that is

$$\mathbb{E}(\mathsf{D}X) = \frac{m}{a+m-1}. \tag{3.13}$$

Reading $I = i$ as the λ-degree carried by the first factor and $J = m - i$ as the degree carried by the second, direct coefficient extraction gives

$$\begin{aligned}[\lambda^m]&\big(ZZ_{a\Theta_\lambda} - Z_aZ_{\Theta_\lambda}\big) \\ &= \sum_{i=0}^{m} \frac{-J\psi(a+J) + J\psi(a+i)}{i!J!\Gamma(a+i)\Gamma(a+J)} = -S_m\mathbb{E}(JX).\end{aligned}$$

Together with $[\lambda^m]Z^2 = S_m$, this yields

$$\frac{[\lambda^m]B}{S_m} = 1 - \mathbb{E}(JX) = 1 - \frac{1}{2}\mathbb{E}(\mathsf{D}X),$$

because $J = (m+\mathsf{D})/2$ and $\mathbb{E}X = 0$ by symmetry. This proves (3.4).

For C_κ, the Euler-derivative terms contribute

$$\kappa \mathbb{E} J - \mathbb{E} J^2 + \mathbb{E}(IJ).$$

Since $I = (m - \mathsf{D})/2$, $J = (m + \mathsf{D})/2$, and $\mathbb{E}\mathsf{D} = 0$ by the symmetry $I \leftrightarrow m - I$,

$$\kappa \mathbb{E} J - \mathbb{E} J^2 + \mathbb{E}(IJ) = \frac{\kappa m}{2} - \frac{1}{2}\mathbb{E}\mathsf{D}^2. \tag{3.14}$$

For $m \geq 2$, cancelling the factors $i(m-i)$ and shifting $j = i - 1$ gives

$$\begin{aligned} S_m(a)\mathbb{E}[I(m-I)] &= \sum_{i=1}^{m-1} \frac{1}{(i-1)!(m-i-1)!\Gamma(a+i)\Gamma(a+m-i)} \\ &= S_{m-2}(a+1). \end{aligned}$$

Therefore

$$\mathbb{E}[I(m-I)] = \frac{S_{m-2}(a+1)}{S_m(a)} = \frac{m(m-1)(a+m-1)}{2(2a+2m-3)},$$

so

$$\mathbb{E}\mathsf{D}^2 = \frac{m(2a+m-2)}{2a+2m-3}.$$

Substitution gives (3.7); the cases $m = 0, 1$ are immediate. □

Remark 3.3 (Finite-law mechanism and the exceptional index)**.** The coefficient calculation has a useful intrinsic interpretation. For fixed m, normalize the convolution weights by

$$\pi_{m,a}(i) = \frac{1}{S_m\, i!(m-i)!\Gamma(a+i)\Gamma(a+m-i)}, \qquad 0 \leq i \leq m,$$

and let $\mathbb{E}_m$ denote expectation with respect to this symmetric finite law, under which I is the random index. Put

$$\mathsf{D} = m - 2I, \qquad X = \psi(a+m-I) - \psi(a+I), \qquad \varrho = \psi_1(a+I) + \psi_1(a+m-I).$$

Then the three normalized coefficient entries may be written as

$$\alpha_m = \tfrac{1}{2}\mathbb{E}_m(\varrho - X^2), \qquad \beta_m = 1 - \tfrac{1}{2}\mathbb{E}_m(\mathsf{D}X), \qquad c_m^{(\kappa)} = \tfrac{\kappa m}{2} - \tfrac{1}{2}\mathbb{E}_m\mathsf{D}^2.$$

The first identity follows by writing

$$\frac{F_m''(0)}{F_m(0)} = \mathbb{E}_m(X^2 - \varrho) = -2\psi_1(a+m),$$

while the other two are the expectation identities obtained in (3.13) and (3.14). Thus the asymmetric Chu–Vandermonde calculation evaluates, respectively, a curvature term, an imbalance–score covariance, and an imbalance variance.

At the sharp endpoint $\kappa = 1$, the last identity is especially revealing:

$$c_m = \tfrac{1}{2}\big(m - \mathbb{E}_m\mathsf{D}^2\big) = \frac{m(m-1)}{2(2a+2m-3)}.$$

Hence $m = 1$ is the only positive degree at which the lower-right variance slack vanishes. This foreshadows the exceptional-index structure of the proof: Theorem 4.2 confirms that M_m is positive definite for every $m \geq 2$, while the mixed-determinant argument shows that the potentially indefinite matrix M_1 can contribute negatively only through its self-pairing, namely in degree $n = 2$. The decomposition of Lemma 5.2 supplies the additional argument required at that single degree.

4. Gram structure and the exceptional matrix M_1

We require elementary two-sided trigamma estimates. The integral representation

$$\psi_1(y) = \int_0^\infty e^{-y\tau} \frac{\tau}{1-e^{-\tau}} \,\mathrm{d}\tau, \qquad y > 0, \tag{4.1}$$

and the series $\psi_1(y) = \sum_{r\geq 0}(y+r)^{-2}$ are standard [16, §5.9 and eq. 5.15.1].

Lemma 4.1 (Trigamma bounds)**.** *For every $y > 0$,*

$$\psi_1(y) > \frac{1}{y} + \frac{1}{2y^2} > \frac{1}{y}. \tag{4.2}$$

For $y > 1/2$,

$$\psi_1(y) < \frac{1}{y-1/2}. \tag{4.3}$$

Consequently,

$$\frac{1}{\psi_1(a)} > a - \frac{1}{2} \qquad (a > 0), \tag{4.4}$$

where the assertion is immediate for $a \leq 1/2$.

Proof. For $f(x) = (y+x)^{-2}$, strict convexity on every interval $[r, r+1]$ gives the strict trapezoidal inequality

$$\int_r^{r+1} f(x)\,\mathrm{d}x < \frac{f(r)+f(r+1)}{2}.$$

Summing and passing to the limit yields

$$\psi_1(y) = \sum_{r\geq 0} f(r) > \int_0^\infty f(x)\,\mathrm{d}x + \frac{1}{2}f(0) = \frac{1}{y} + \frac{1}{2y^2},$$

which proves (4.2). For the upper bound, the elementary inequality $\tau < 2\sinh(\tau/2)$ for $\tau > 0$ is equivalent to $\tau/(1-e^{-\tau}) < e^{\tau/2}$. Substitution in (4.1) gives (4.3). Taking reciprocals proves (4.4) when $a > 1/2$. □

Fix $m \geq 1$ and write

$$x = a + m, \qquad q = a + \frac{m}{2} - 1.$$

In $\ell^2(\mathbb{N}_0)$ define

$$u_r^{(m)} = \frac{1}{x+r}, \qquad v_r^{(m)} = \frac{q}{x+r-1}, \qquad r \geq 0.$$

Then

$$\|u^{(m)}\|^2 = \psi_1(x) = \alpha_m, \tag{4.5}$$

$$\langle u^{(m)}, v^{(m)}\rangle = q\sum_{r\geq 0}\frac{1}{(x+r-1)(x+r)} = \frac{q}{x-1} = \beta_m, \tag{4.6}$$

and

$$\|v^{(m)}\|^2 = q^2\psi_1(x-1). \tag{4.7}$$

Define

$$\rho_m(a) = \frac{1}{g} + c_m - q^2\psi_1(x-1). \tag{4.8}$$

Call the coefficient matrix M_m *stable* when $m \geq 2$, or $m = 1$ and $a \geq 1/2$. For vectors x, y in a Hilbert space, $\mathrm{Gram}(x, y)$ denotes the symmetric 2×2 matrix with entries $\langle x, x\rangle$, $\langle x, y\rangle$, $\langle y, y\rangle$. Write $N_m = \mathrm{diag}(1, g^{-1/2})M_m\,\mathrm{diag}(1, g^{-1/2})$ for the normalized coefficient matrix.

Theorem 4.2 (Gram representation of stable coefficients)**.** *If either $m \geq 2$ and $a > 0$, or $m = 1$ and $a \geq 1/2$, then $\rho_m(a) > 0$. In $\ell^2(\mathbb{N}_0) \oplus \mathbb{R}$, put*

$$\xi_m = (u^{(m)}, 0), \qquad \eta_m = (v^{(m)}, \sqrt{\rho_m(a)}).$$

Then

$$N_m = \mathrm{Gram}(\xi_m, \eta_m), \tag{4.9}$$

$$M_m = \mathrm{Gram}(\xi_m, \sqrt{g}\,\eta_m). \tag{4.10}$$

In particular, M_m is positive definite in these ranges.

Proof. Assume first that $m \geq 2$. Then $x - 1 > 1/2$, so Lemma 4.1 gives

$$\rho_m(a) > a - \frac{1}{2} + c_m - \frac{q^2}{x - 3/2}.$$

At the sharp endpoint the right-hand side simplifies exactly to

$$a - \frac{1}{2} + \frac{m(m-1)}{2(2a + 2m - 3)} - \frac{(a + m/2 - 1)^2}{a + m - 3/2} = \frac{m-1}{2(2a + 2m - 3)} > 0.$$

For $m = 1$, $q = a - 1/2$ and $c_1 = 0$. If $a > 1/2$, then

$$q^2 \psi_1(a) < q < \frac{1}{\psi_1(a)},$$

by (4.3) and (4.4). Hence $\rho_1(a) > 0$. At $a = 1/2$, $q = 0$ and the conclusion is immediate. The Gram identities now follow from (4.5)–(4.8). ☐

The constant matrix is

$$M_0 = \begin{pmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{g} \\ 1 \end{pmatrix} \begin{pmatrix} \sqrt{g} & 1 \end{pmatrix},$$

a rank-one positive-semidefinite matrix. With $\psi_1(a+1) = g - a^{-2}$ and $\beta_1 = (2a-1)/(2a)$, the remaining coefficient matrix M_1 has determinant

$$\det M_1 = \psi_1(a+1) - g\left(\frac{2a-1}{2a}\right)^2 = \frac{(4a-1)g - 4}{4a^2}. \tag{4.11}$$

Remark 4.3 (Genuine indefiniteness of M_1)**.** The anomaly at $m = 1$ is genuine. Set $f(a) = (4a-1)\psi_1(a) - 4$, so that $\det M_1$ has the sign of $f(a)$ by (4.11). Since $\psi_1(a) = \sum_{r \geq 0} (a+r)^{-2}$, for $0 < a < 1/2$

$$f'(a) = 2 \sum_{r \geq 0} \frac{2r + 1 - 2a}{(a+r)^3} > 0,$$

while $f(a) \to -\infty$ as $a \downarrow 0$ and $f(1/2) = \psi_1(1/2) - 4 = \pi^2/2 - 4 > 0$. Hence f has a unique zero $a_* \in (0, 1/2)$. With Theorem 4.2 for $a \geq 1/2$ and the positive diagonal of M_1, the matrix M_1 is indefinite for $0 < a < a_*$, positive semidefinite of rank one at $a = a_*$, and positive definite for $a > a_*$. It is the only coefficient matrix that can be indefinite at the endpoint $\kappa = 1$, so the determinant positivity of Theorem 2.2 cannot follow from pointwise semidefiniteness of the M_m alone.

5. Mixed determinants and coefficientwise positivity

For symmetric 2×2 matrices $X = (x_{ij})$ and $Y = (y_{ij})$, define

$$\mathrm{MD}(X, Y) = x_{11}y_{22} + x_{22}y_{11} - 2x_{12}y_{12}.$$

This is the polarization of the determinant.

Lemma 5.1 (Wedge-square positivity)**.** *If $X, Y \succeq 0$, then $\mathrm{MD}(X, Y) \geq 0$. If $X \neq 0$ and $Y \succ 0$, then $\mathrm{MD}(X, Y) > 0$.*

Proof. Write $X = \sum_r x_r x_r^T$ and $Y = \sum_s y_s y_s^T$. Then

$$\mathrm{MD}(X, Y) = \sum_{r,s} (x_{r,1} y_{s,2} - x_{r,2} y_{s,1})^2 .$$

If this sum vanished and $X \neq 0$, fix a nonzero Gram vector x_r. Every Gram vector y_s would then be parallel to x_r, forcing Y to have rank at most one, contrary to $Y \succ 0$. $\square$

Expanding the determinant of (3.8) gives

$$\Delta_n = \frac{1}{2} \sum_{k=0}^{n} S_k S_{n-k} \, \mathrm{MD}(M_k, M_{n-k}). \tag{5.1}$$

For $a \geq 1/2$, M_0 is rank-one positive semidefinite and $M_m \succ 0$ for $m \geq 1$ by Theorem 4.2, so Lemma 5.1 makes every summand of (5.1) nonnegative and the pair (M_0, M_n) strictly positive, giving $\Delta_n > 0$; the argument that follows is uniform in $a > 0$. The linear coefficient is strictly positive: (4.2) gives $\psi_1(a) > 1/a$, so $ag > 1$ and

$$\mathrm{MD}(M_0, M_1) = \frac{ag - 1}{a^2} > 0,$$

whence

$$\Delta_1 = S_0 S_1 \, \mathrm{MD}(M_0, M_1) = \frac{2(ag - 1)}{a^3 \Gamma(a)^4} > 0. \tag{5.2}$$

For $m \geq 2$ the mixed term satisfies

$$\mathrm{MD}(M_1, M_m) = \alpha_1 (1 + g c_m) + \alpha_m - 2 g \beta_1 \beta_m \geq 0. \tag{5.3}$$

If $a \geq 1/2$ then $M_1, M_m \succeq 0$ by Theorem 4.2 and the nonstrict case of Lemma 5.1 applies; if $0 < a < 1/2$ then $\beta_1 < 0 < \beta_m$ by (3.4), and since every diagonal entry is positive the right-hand side is strictly positive. For $n \geq 3$ the self-pair $(1, 1)$ does not occur. Classify the pairs $(k, n-k)$ in (5.1): if neither index is 1, both matrices in the pair are positive semidefinite—M_0 is rank-one positive semidefinite and $M_m \succ 0$ for $m \geq 2$ by Theorem 4.2—so the nonstrict case of Lemma 5.1 gives a nonnegative summand; if one index is 1, the other is at least 2 and (5.3) applies. Every summand is therefore nonnegative, and the pair (M_0, M_n) is strictly positive because $M_0 \neq 0$ and $M_n \succ 0$. Hence $\Delta_n > 0$. The self-pairing $\mathrm{MD}(M_1, M_1) = 2 \det M_1$, excluded by this classification, occurs solely in Δ_2; by Remark 4.3 the indefiniteness of M_1 for small $a > 0$ can contribute negatively only there.

5.1. The exceptional determinant coefficient Δ_2.

Lemma 5.2 (Exceptional degree-two decomposition)**.** *For every $a > 0$, one has $\Delta_2(a) > 0$. More precisely, let*

$$t = \psi_1(a + 1), \qquad \mathcal{R}_a = t - \frac{1}{a + 1}.$$

Then

$$\Delta_2 = \frac{Q_*(a, t)}{2 a^6 (a + 1)^3 \Gamma(a)^4}, \tag{5.4}$$

where

$$\begin{aligned} Q_*(a, t) = {} & 2a^4 (a + 1)^2 \mathcal{R}_a^2 \\ & + 2a^2 (a + 1)^2 (8a^2 + 3a + 1) \mathcal{R}_a \\ & + 2a(a + 1)(5a + 3). \end{aligned} \tag{5.5}$$

Proof. From (5.1),

$$\Delta_2 = S_0 S_2 \, \mathrm{MD}(M_0, M_2) + S_1^2 \det M_1. \tag{5.6}$$

Here

$$S_0 = \frac{1}{\Gamma(a)^2}, \qquad S_1 = \frac{2}{a \Gamma(a)^2}, \qquad S_2 = \frac{2a + 1}{a^2 (a + 1) \Gamma(a)^2},$$

and, recalling $\det M_1 = t - g((2a-1)/(2a))^2$ from (4.11),

$$\mathrm{MD}(M_0, M_2) = g\left(1 + \frac{g}{2a+1}\right) + \psi_1(a+2) - 2g\frac{a}{a+1}.$$

Substituting $g = t + a^{-2}$ and $\psi_1(a+2) = t - (a+1)^{-2}$ into (5.6) and collecting over the denominator in (5.4) first gives

$$\begin{aligned} Q_*(a,t) = {} & 2a^4(a+1)^2t^2 \\ & + 2a^2(8a^4 + 17a^3 + 13a^2 + 5a + 1)t \\ & + 2a(-8a^4 - 10a^3 + a^2 + 7a + 3). \end{aligned}$$

After multiplication by the common denominator in (5.4), the preceding formula is a polynomial identity in (a,t). Replacing t by $\mathcal{R}_a + (a+1)^{-1}$ and collecting this polynomial yields the identity (5.5).

By (4.1),

$$\mathcal{R}_a = \int_0^\infty e^{-(a+1)s}\left(\frac{s}{1-e^{-s}} - 1\right) \mathrm{d}s > 0.$$

All three coefficients in (5.5) are positive for $a > 0$, so $\Delta_2 > 0$. □

Proof of Theorem 2.2. The preceding argument, together with Lemma 5.2, proves the endpoint assertion $\Delta_n^{(1)}(a) > 0$ for every $a > 0$ and $n \geq 1$. Since A and Z have positive coefficients and

$$[\lambda^n]Z_{\Theta_\lambda} = \frac{n}{n!\Gamma(a+n)} > 0 \qquad (n \geq 1),$$

every positive-degree coefficient of AZZ_{Θ_λ} is strictly positive: in degree n, for example, the product of the constant coefficients of A and Z with $[\lambda^n]Z_{\Theta_\lambda}$ is positive. Therefore

$$\Delta^{(\kappa)} = \Delta^{(1)} + g(\kappa - 1)A\,ZZ_{\Theta_\lambda}$$

shows that all coefficients remain strictly positive for $\kappa > 1$.

For the converse, the general $m = 1$ coefficient gives

$$\mathrm{MD}(M_0, M_1^{(\kappa)}) = \frac{\frac{\kappa-1}{2}(ag)^2 + ag - 1}{a^2}. \tag{5.7}$$

If $\kappa < 1$, then $ag \to +\infty$ as $a \downarrow 0$, so (5.7), and hence $\Delta_1^{(\kappa)}$, is negative for all sufficiently small $a > 0$. This proves the sharp classification.

Finally, $\Delta(a,\cdot)$ is entire, so each differentiated Maclaurin series converges locally uniformly. Hence, for every $r \geq 0$ and $\lambda > 0$,

$$\partial_\lambda^r \Delta(a,\lambda) = \sum_{n \geq \max\{1,r\}} \frac{n!}{(n-r)!}\Delta_n(a)\lambda^{n-r} > 0.$$

Thus $\Delta(a,\cdot)$ is strictly absolutely monotone. Also $A > 0$ for $\lambda > 0$ by (3.2) and (3.3). Hence $\Delta > 0$ and the leading principal entry is positive, so $\mathcal{T} \succ 0$. □

Combining (5.2) with (4.2), which gives $ag - 1 > (2a)^{-1}$, yields the explicit lower bound

$$\Delta(a,\lambda) > \Delta_1(a)\lambda = \frac{2\big(a\psi_1(a) - 1\big)}{a^3\Gamma(a)^4}\lambda > \frac{\lambda}{a^4\Gamma(a)^4} \qquad (a, \lambda > 0).$$

6. Exact reduction from ${}_0F_1$ to the Bessel inequality

The standard identity [15, eq. 10.25.2], [21, §3.7]

$$I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2}Z(a,\lambda) \tag{6.1}$$

with $L = \log Z$, $z = 2\sqrt{\lambda}$, and $\nu = a - 1$ gives

$$\Theta_z \log I_\nu = \nu + 2L_{\Theta_\lambda}, \qquad P_\nu = 1 + 2L_{a\Theta_\lambda}, \tag{6.2}$$

$$G_\nu = -L_{aa}, \qquad H_\nu = 4(L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}). \tag{6.3}$$

Moreover,

$$\frac{A}{Z^2} = -L_{aa}, \qquad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \qquad \frac{C}{gZ^2} = \frac{1}{g} + L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}. \tag{6.4}$$

The Bessel–Schur matrix is therefore the congruence

$$\begin{pmatrix} G_\nu & 1+P_\nu \\ 1+P_\nu & H_\nu + 4/g \end{pmatrix} = \frac{1}{Z^2} \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix} \mathcal{T}(a,\lambda) \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix}, \tag{6.5}$$

whose determinant is

$$D_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta(a,\lambda)}{gZ(a,\lambda)^4}. \tag{6.6}$$

This reduces Theorem 2.4 and Corollary 2.5 to the endpoint $\kappa = 1$ of Theorem 2.2; the proofs are completed in Section 7.

7. Bessel consequences and sharpness

The coefficientwise theorem of Section 5, transported through the exact reduction of Section 6, gives the sharp Bessel inequality; the same computation fixes its optimal constants, and its continuation past the positive-series domain exhibits the negative-order failure.

Proof of Theorem 2.4 and Corollary 2.5. The congruence (6.5) and Theorem 2.2 give positive definiteness for $z > 0$, and (6.6) gives the determinant inequality. As $z \downarrow 0$, equivalently $\lambda \downarrow 0$, the power series for $L = \log Z$ gives $L_{\Theta_\lambda}, L_{a\Theta_\lambda}, L_{\Theta_\lambda\Theta_\lambda} \to 0$. Hence (6.2)–(6.3) imply

$$G_\nu \longrightarrow g, \qquad 1 + P_\nu \longrightarrow 2, \qquad H_\nu + \frac{4}{g} \longrightarrow \frac{4}{g},$$

which is the matrix limit stated in Corollary 2.5. $\square$

Proof of Proposition 2.6. For the least z-independent correction, replace $4/\psi_1(\nu+1)$ in (2.13) by a z-independent $R(\nu)$ and write $D_{\nu,R}$ for the resulting determinant. Then

$$D_{\nu,R}(z) \longrightarrow \psi_1(\nu+1)R(\nu) - 4 \qquad (z \downarrow 0),$$

so validity for every $z > 0$ forces $R(\nu) \geq 4/\psi_1(\nu+1)$. Conversely $D_{\nu,R} = D_\nu + G_\nu\big(R(\nu) - 4/\psi_1(\nu+1)\big) > 0$ whenever $R(\nu) \geq 4/\psi_1(\nu+1)$, since $D_\nu > 0$ by Theorem 2.4 and $G_\nu = A/Z^2 > 0$ for $\nu > -1$, $z > 0$. Hence $4/\psi_1(\nu+1)$ is the exact least z-independent correction.

For the fixed-correction deformation (2.11)–(2.12), the same reduction gives

$$D^{(\kappa)}_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta^{(\kappa)}(a,\lambda)}{gZ(a,\lambda)^4}. \tag{7.1}$$

A direct first-order expansion,

$$Z(a,\lambda) = \frac{1}{\Gamma(a)}\left(1 + \frac{\lambda}{a} + O(\lambda^2)\right),$$

together with the corresponding termwise parameter and Euler derivatives in (2.2)–(2.4), gives

$$\Delta^{(\kappa)}(a,\lambda) = \frac{2}{a^3\Gamma(a)^4}\left[\frac{\kappa-1}{2}(ag)^2 + ag - 1\right]\lambda + O(\lambda^2).$$

Using (7.1) and $\lambda = z^2/4$ therefore yields

$$D^{(\kappa)}_{a-1}(z) = \frac{2}{a^3 g}\left[\frac{\kappa-1}{2}(ag)^2 + ag - 1\right] z^2 + O(z^4). \tag{7.2}$$

For $\kappa < 1$, the bracket is negative for all sufficiently small $a > 0$, because $ag \to +\infty$ as $a \downarrow 0$, proving necessity of $\kappa \geq 1$; sufficiency follows from Theorem 2.2 through (7.1). $\square$

7.1. **Continuation beyond the positive-series domain.** The behavior changes qualitatively once the order crosses the endpoint $\nu = -1$ of the positive-series domain. On the open set

$$\mathcal{U} = \{(\nu, z) \in (\mathbb{R} \setminus \{-1, -2, \ldots\}) \times (0, \infty) : I_\nu(z) \neq 0\},$$

define G_ν, P_ν, H_ν, and D_ν by (2.7)–(2.10) with $\log I_\nu$ replaced by $\log |I_\nu(z)|$; the order derivatives are taken on $\mathcal{U}$, and for each fixed ν the resulting formulas restrict to any connected component of $\{z > 0 : I_\nu(z) \neq 0\}$. Where $I_\nu > 0$ this agrees with the ordinary logarithm.

For every real $a \notin \{0, -1, -2, \ldots\}$, the partial-fraction expansion

$$\psi_1(a) = \sum_{k=0}^{\infty} \frac{1}{(a+k)^2} \tag{7.3}$$

holds and shows that $g = \psi_1(a) > 0$ [16, eq. 5.15.1]. Hence the positive real square root $\sqrt{g}$ occurring in (2.5) and (6.5) remains well defined on every real nonpole component.

Lemma 7.1 (Meromorphic continuation of the coefficient identities)**.** *For each fixed coefficient index, the scalar coefficient identities in* (3.2)*–*(3.7)*, and every resulting finite-convolution formula for a coefficient of* Δ*, extend meromorphically in* a*. This assertion concerns the scalar entries and* Δ*; it does not require a single-valued meromorphic choice of* $\sqrt{\psi_1(a)}$ *on the complex plane. If* $a \in \mathbb{R} \setminus \{0, -1, -2, \ldots\}$*,* $\lambda > 0$*, and* $Z(a, \lambda) \neq 0$*, then* (6.4)*,* (6.5)*, and* (6.6) *remain valid, with the positive real square root of* $\psi_1(a)$ *and with logarithmic derivatives interpreted through* $\log |Z|$ *and* $\log |I_{a-1}|$ *on the relevant nonvanishing component.*

Proof. Each summand of (2.1) is entire in a. Uniform Stirling estimates for a in compact sets, together with Cauchy estimates for the derivatives of $1/\Gamma$, give locally uniform convergence of the series and of every fixed-order parameter and Euler derivative on compact subsets of $\mathbb{C}^2$; hence Z and these derivatives are entire in (a, λ). For each fixed coefficient index, both sides of each scalar coefficient identity are meromorphic functions of the single variable a and agree on $a > 0$; since $a > 0$ has a limit point, the one-variable identity theorem shows they agree throughout their common domain, the complement in $\mathbb{C}$ of the discrete pole set. Each coefficient of $\Delta = AC - gB^2$ is a finite convolution of such scalar coefficient functions and is therefore meromorphic as well.

On the real nonpole set, (7.3) gives $g = \psi_1(a) > 0$, so the positive real square root used in the matrix formulas is well defined. Let

$$\mathcal{V} = \{(a, \lambda) : a \notin \{0, -1, -2, \ldots\},\ \lambda > 0,\ Z(a, \lambda) \neq 0\}.$$

On each connected component of $\mathcal{V}$, Z has constant sign, so $\partial_a \log |Z| = Z_a/Z$ and the corresponding mixed and second derivative identities hold there. Finally, (6.1) gives $\log |I_{a-1}(2\sqrt{\lambda})| = \frac{a-1}{2} \log \lambda + \log |Z(a, \lambda)|$ where $Z \neq 0$, leaving the derivations of (6.4), (6.5), and (6.6) unchanged on that component. $\square$

Away from the poles the leading small-z term keeps its positive sign. For $a = \nu + 1 \in \mathbb{R} \setminus \{0, -1, -2, \ldots\}$ the ascending series gives $I_\nu(z) = (z/2)^\nu \Gamma(\nu+1)^{-1}(1 + O(z^2))$, so I_ν has constant sign near $z = 0$; by Lemma 7.1, (5.2) and (6.6) with $Z(a, 0) = 1/\Gamma(a)$ and $\lambda = z^2/4$ give

$$D_\nu(z) = \frac{2\big(a\psi_1(a) - 1\big)}{a^3 \psi_1(a)} z^2 + O(z^4) \qquad (z \downarrow 0), \tag{7.4}$$

whose coefficient is positive for every such a: for $a > 0$ by (4.2), and for negative nonintegral a because (7.3) makes $a\psi_1(a) - 1$ and $a^3\psi_1(a)$ both negative. Coefficientwise positivity of Δ nonetheless fails already on the next interval where $1/\Gamma(a) > 0$: for $-3 < \nu < -2$ one has $a \in (-2, -1)$ and $\Gamma(a) > 0$.

Proposition 7.2 (Degree-two obstruction beyond the positive-series domain)**.** *If* $-2 < a < -1$*, equivalently* $-3 < \nu < -2$*, then*

$$[\lambda^2]\Delta(a, \lambda) = \Delta_2(a) < 0. \tag{7.5}$$

Thus the coefficientwise positivity theorem for Δ *cannot be extended to that interval.*

Proof. By Lemma 7.1, the identity (5.4)–(5.5) persists on $-2 < a < -1$. Set $b = -a \in (1, 2)$. Since $a+1 \in (-1, 0)$, (7.3) gives $t = \psi_1(a+1) > 0$,

$$\mathcal{R}_a = t - \frac{1}{a+1} > \frac{1}{b-1}.$$

The first term in (5.5) is positive. The second and third terms satisfy

$$\begin{aligned} &2b^2(b-1)^2(8b^2 - 3b + 1)\mathcal{R}_a - 2b(b-1)(5b-3) \\ &\quad > 2b(b-1)\left[8b^3 - 3b^2 - 4b + 3\right] > 0. \end{aligned}$$

Indeed, the polynomial in brackets equals 4 at $b = 1$ and has derivative $24b^2 - 6b - 4 > 0$ for $b \geq 1$. Hence $Q_*(a, t) > 0$. But $2a^6(a+1)^3\Gamma(a)^4 < 0$, proving (7.5). $\square$

Remark 7.3 (Negative-order features)**.** The continued determinant therefore has two distinct negative-order features: positive leading small-z behavior away from the poles, yet failure of coefficientwise positivity of Δ on all of $-3 < \nu < -2$. A complete classification beyond $\nu > -1$ appears to require separate treatment.

8. Comparison with related positivity results

The scalar quantity $A = Z_a^2 - ZZ_{aa}$ is the infinitesimal symmetric-shift Turánian of the reciprocal-gamma series. Indeed, $Z(a+\delta, \lambda)Z(a-\delta, \lambda) - Z(a, \lambda)^2 = -A(a, \lambda)\delta^2 + O(\delta^4)$. The scalar coefficient sign of A is the infinitesimal case of the coefficientwise reciprocal-gamma Turán inequalities of Kalmykov and Karp [9, Thm. 3.1 and Ex. 1] (the integer-shift form in [12, Lem. 1 and Thm. 3]), whose gamma-ratio results similarly treat scalar two-function Turánians under parameter shifts [10, 11]. The sharp coefficientwise thresholds of Karp and Zhang [13, §3] are set by the multiplicity of Pochhammer symbols in such scalar series; the threshold here is instead the deformation parameter κ of a coupled matrix determinant.

The additional quantities B and C are not scalar parameter-shift Turánians: B contains a mixed parameter–Euler derivative, while C contains the sharp argument-curvature expression and the trigamma normalization that becomes the least z-independent correction under the Bessel congruence. The result proved here is the coefficientwise positivity of the coupled determinant $AC - \psi_1(a)B^2$, even though the coefficient matrix M_1 can be indefinite.

Classical Bessel Turán inequalities compare neighboring orders, for example $I_\nu^2 - I_{\nu-1}I_{\nu+1}$, or control ratios and logarithmic derivatives [1, 2, 4, 20]; series representations of the modified-Bessel Turánian [14, Thms. 1 and 5], positivity of Turán determinants for symmetric orthogonal polynomials [6, Thm. 1 and Cor. 1], and higher-order Turán and Hankel determinants in the order [3, §§2.2, 3] have also been studied, as has the monotonicity in the argument of determinants with special-function entries [8, Thm. 2.3]. In contrast, (2.13) couples second curvature in the order with first mixed order–argument curvature and second argument curvature. First-order derivatives of I_ν with respect to the order, which underlie G_ν and P_ν, have been studied in their own right, with closed forms and integral representations [17, §10.38], [7]; those results represent the separate order derivatives rather than the mixed-curvature determinant proved positive here.

A recent preprint of Salazar establishes that the kernel $(x, s) \mapsto I_s(x)$ is strictly totally positive for $x > 0$ and $s \geq 0$ [18], extending integer-order results of Buchstaber and Glutsyuk [5, Thm. 1.4]; a companion preprint builds all-order Hankel and Turán determinant hierarchies for special functions from positive moment representations [19]. Their determinant hierarchies are respectively of evaluation-minor and Hankel-moment type; none of these papers states or proves the coefficientwise mixed-derivative determinant studied here. The signs of all total-positivity minors of $(x, s) \mapsto I_s(x)$, including their confluent limits, are preserved under positive row and column rescalings of the kernel, whereas the Schur determinant of (2.13) is not. To make this explicit, write

$$F(s, y) = \log I_s(e^y), \qquad p = F_{sy}, \qquad G = -F_{ss},$$

and replace $I_s(e^y)$ by $e^{cy}I_s(e^y)$, so that $\widetilde{F} = F + cy$ has

$$\widetilde{G} = -\widetilde{F}_{ss} = G, \qquad \widetilde{p} = \widetilde{F}_{sy} = p, \qquad \widetilde{H} = 2(\widetilde{F}_y - s) - \widetilde{F}_{yy} = H + 2c.$$

Consequently, writing the Schur determinant as

$$D_s(e^y) = G\left(H + \frac{4}{\psi_1(s+1)}\right) - (1+p)^2,$$

the rescaled kernel has

$$\widetilde{D}_s(e^y) = D_s(e^y) + 2cG,$$

although all total-positivity signs are unchanged.

9. Further questions

Question 9.1 (Wright generalization)**.** For $\rho > 0$, let

$$Z_\rho(a,\lambda) = \sum_{k\ge 0} \frac{\lambda^k}{k!\Gamma(a+\rho k)}, \qquad L_\rho = \log Z_\rho,$$

and, for $\kappa \in \mathbb{R}$, define

$$\frac{A_\rho}{Z_\rho^2} = -\partial_a^2 L_\rho, \qquad \frac{B_\rho}{Z_\rho^2} = 1 + \rho\,\partial_a \Theta_\lambda L_\rho,$$

$$\frac{C_{\rho,\kappa}}{\psi_1(a) Z_\rho^2} = \frac{1}{\psi_1(a)} + \kappa\rho\,\Theta_\lambda L_\rho - \rho^2 \Theta_\lambda^2 L_\rho, \qquad \Delta_{\rho,\kappa} = A_\rho C_{\rho,\kappa} - \psi_1(a) B_\rho^2.$$

For which pairs (ρ,κ) does $\Delta_{\rho,\kappa}$ have strictly positive Maclaurin coefficients in every positive degree for all $a > 0$; and for each $\rho > 0$, what is the sharp threshold in κ? At $\rho = 1$ the present theorem gives exactly $\kappa \ge 1$, and its proof makes essential use of the exact $\rho = 1$ Chu–Vandermonde convolution.

Question 9.2 (Structural origin)**.** Is there a canonical normalization, bordered kernel, or nonlocal minor associated with $(x,\nu) \mapsto I_\nu(x)$ whose positivity is equivalent to the Bessel–Schur inequality of Theorem 2.4?

The gauge obstruction above shows that any such derivation must incorporate additional Bessel-specific normalization or nonlocal information. A related question is whether the coefficientwise positivity of $AC - \psi_1(a)B^2$ follows from an appropriately strengthened form of scalar reciprocal-gamma log-concavity.

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